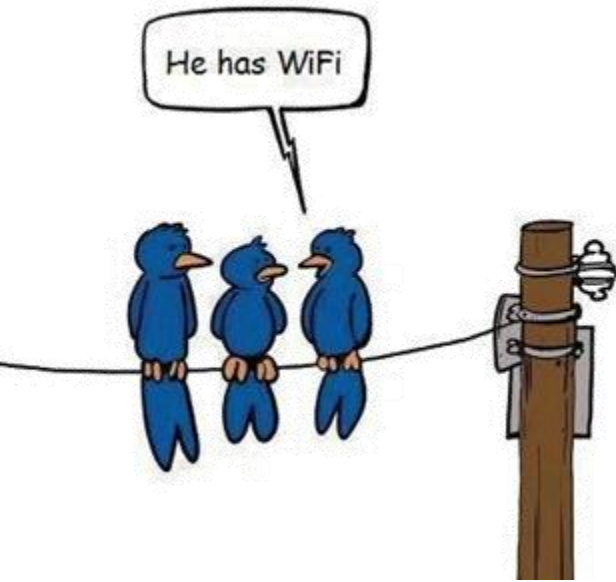


ECSE 4560/ 6560

Modern Communication Systems

Lecture 13: OFDM SFO

Ish Jain, Fall 2025



Announcements

- Join Piazza: <https://piazza.com/rpi/fall2025/ecse45606560>
- Per-Class Quiz
 - Posted on LMS; Due Today 11:59 PM
- Attendance Link: <https://forms.office.com/r/p6P9YrN1d8>
 - Today's Magic Word: **SFO**
- Projects
 - Group formation [MCS_Fall25_project.xlsx](#)
- HW4 will be posted soon.
- Exam 1: Friday Oct 24
- Exam 2: Friday Nov 21
- Final Project Presentation and Demos: Dec 9 and Dec 12
- Project Report and peer evaluation: Finals exam week

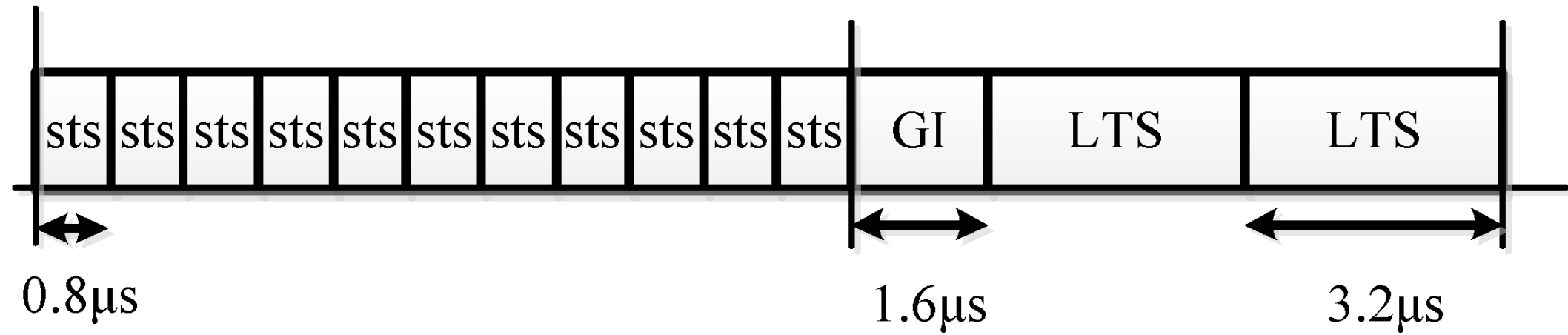
Extra Reading on Piazza

- Principles of Modern Wireless Communication Systems Theory and Practice; by Aditya K. Jagannatham, IIT Kanpur Book
 - Parts of Chapter 1,2,3,4,7
- An Adaptive Orthogonal Frequency Division Multiplexing Baseband Modem for Wideband Wireless Channels; by Jit Ken Tan, MIT Thesis
 - OFDM Decoding and Synchronization
 - Lab 4 Implementation

Today's Class: OFDM SFO

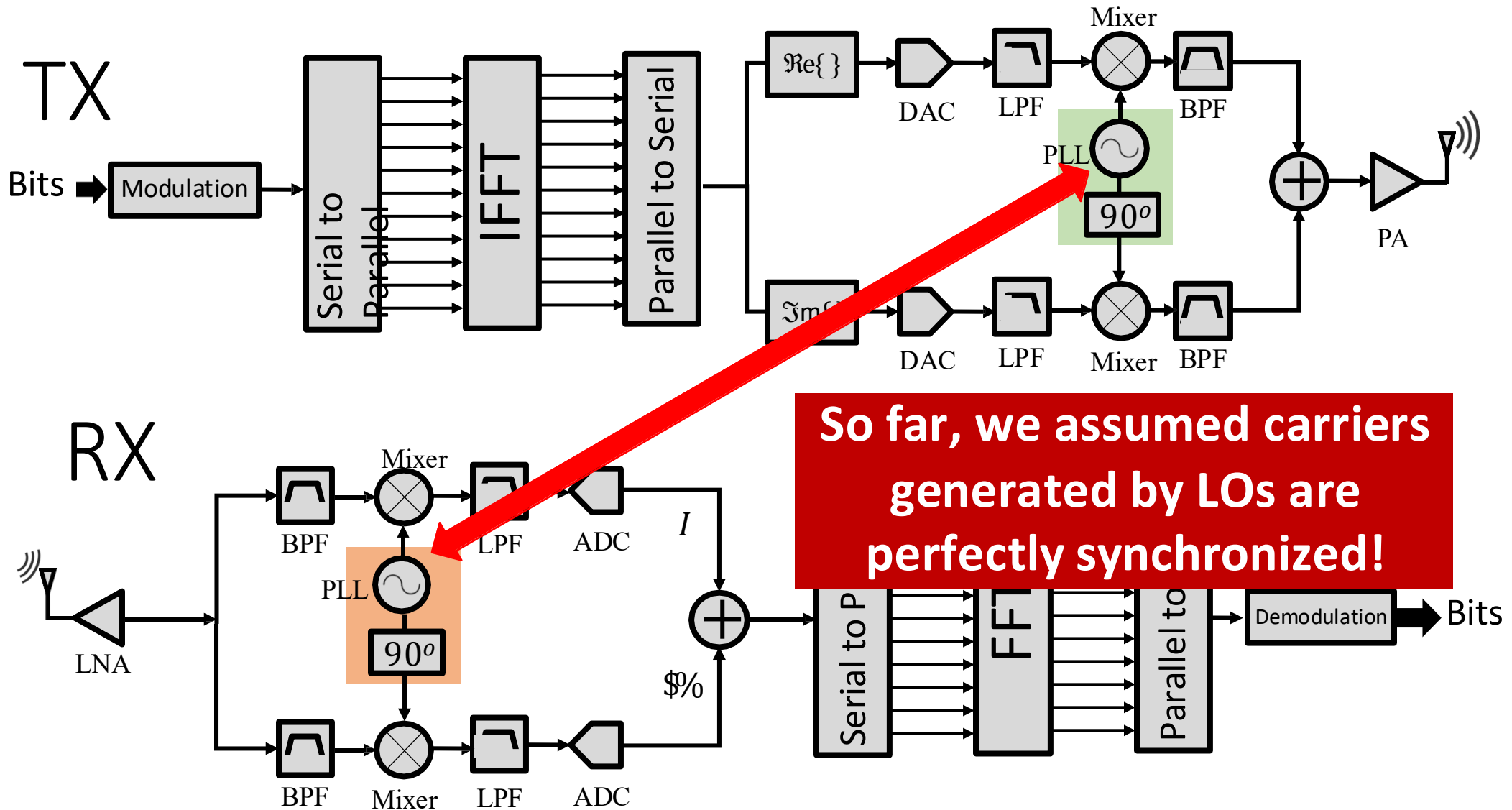
- OFDM Synchronization
 - CFO correction (Revise)
 - Channel Estimation (Revise)
 - SFO and Residual Phase correction

WiFi Example: STS and LTS

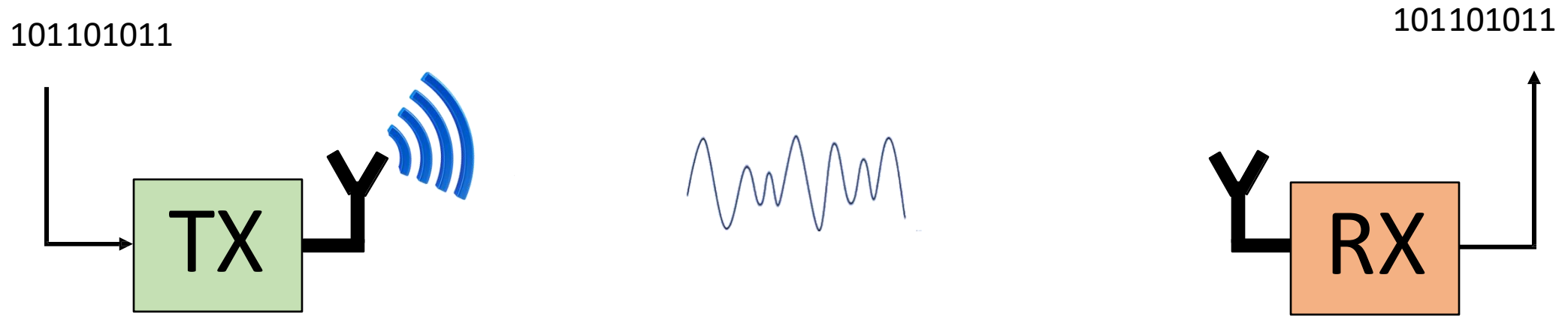


OFDM: Orthogonal Frequency Division

Multiplexing Transmit Symbols in Frequency Domain On Orthogonal Subcarriers



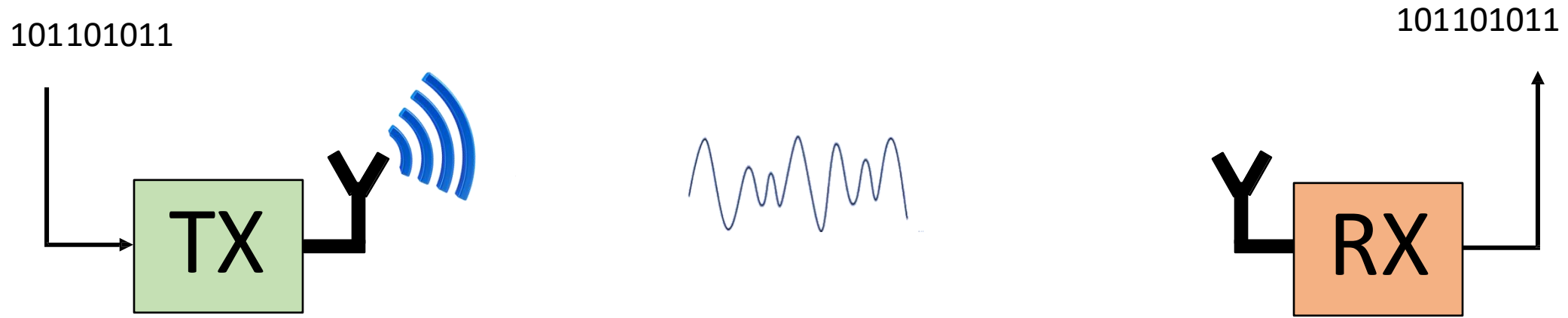
Carrier Frequency Offset



$$\begin{aligned}
 x(t) &\Rightarrow x(t) \times e^{-j2\pi f_c t} \Rightarrow h(t) * x(t) e^{-j2\pi f_c t} \Rightarrow h(t) * x(t) e^{-j2\pi f_c t} \times e^{j2\pi f_c t} \\
 &\Rightarrow h(t) * x(t) \\
 &\Rightarrow y(t) = h(t) * x(t) + v(t)
 \end{aligned}$$

Assumes TX & RX perfectly synched

Carrier Frequency Offset



$$x(t) \Rightarrow x(t) \times e^{-j2\pi f_c t} \Rightarrow h(t) * x(t) e^{-j2\pi f_c t} \Rightarrow h(t) * x(t) e^{-j2\pi f_c t} \times e^{j2\pi f'_c t}$$

TX & RX are not synched

$$\Rightarrow h(t) * x(t) e^{-j2\pi \Delta f_c t}$$

$$\text{CFO: } \Delta f_c = f_c - f'_c$$

$$\Rightarrow y(t) = h(t) * x(t) e^{-j2\pi \Delta f_c t} + v(t)$$

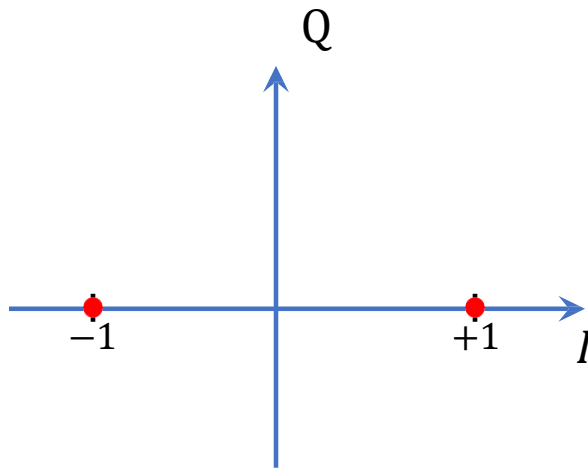
Phase changes with time!

Carrier Frequency Offset

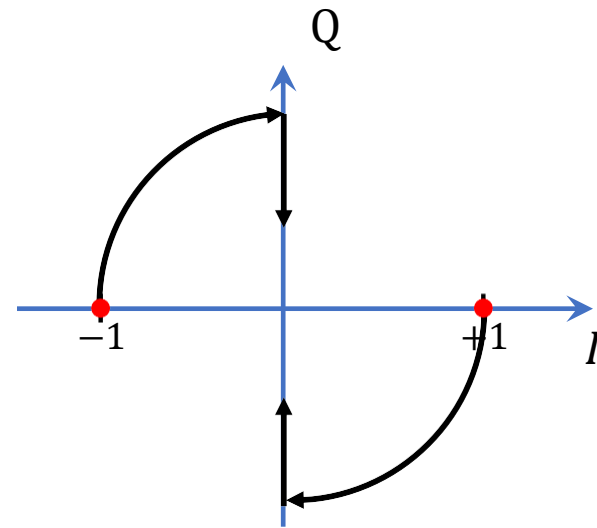
Consider BPSK Modulation.

$0 \rightarrow -1$

$1 \rightarrow +1$



$x(t)$



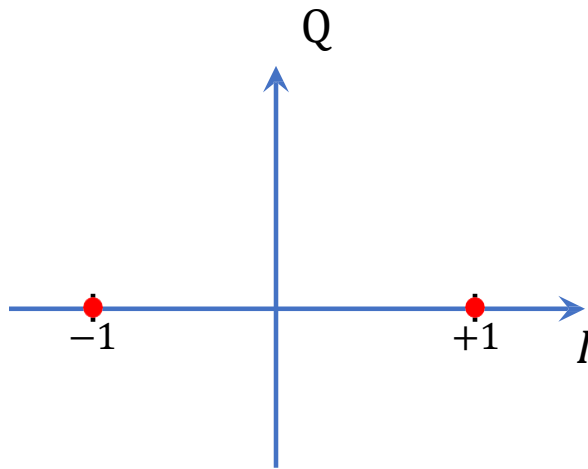
$h x(t - r) e^{-j2\pi\Delta f_c t} + v(t)$

Carrier Frequency Offset

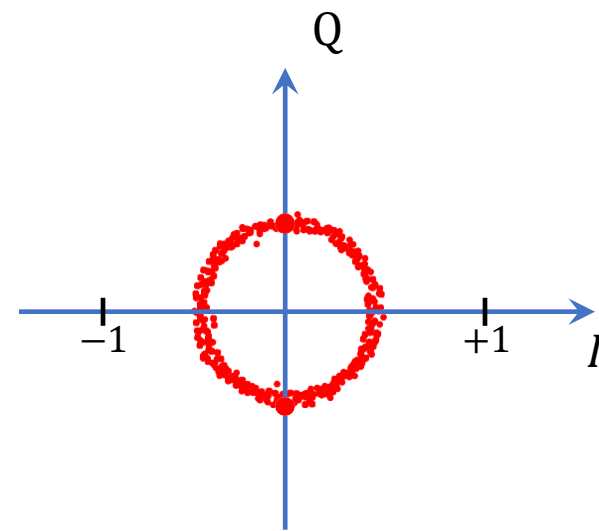
Consider BPSK Modulation.

$0 \rightarrow -1$

$1 \rightarrow +1$



$x(t)$

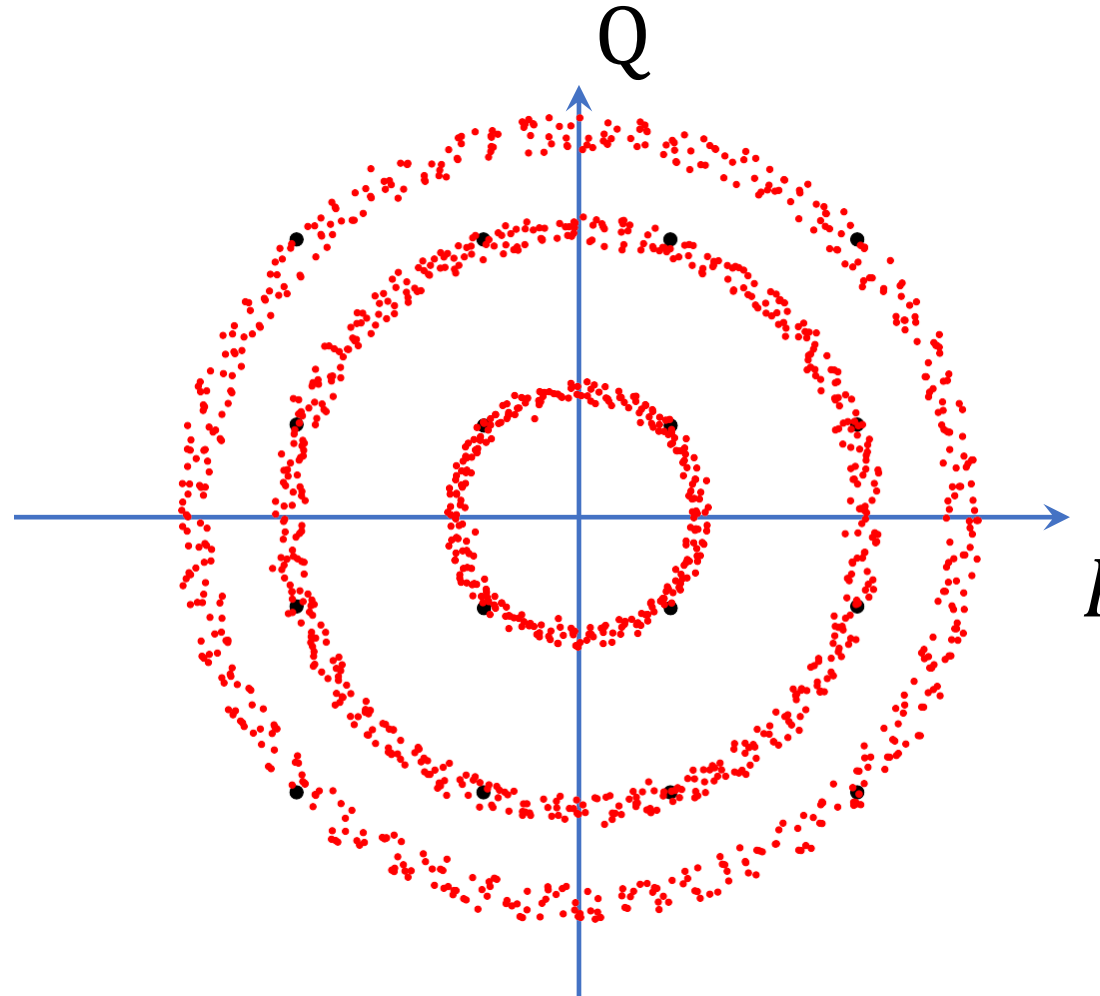


$h x(t - r) e^{-j2\pi\Delta f_c t} + v(t)$

Impossible to Decode!

Carrier Frequency Offset

Consider 16 QAM Modulation



Need to estimate and correct CFO to decode!

OFDM CFO Estimation & Correction



- Use Preamble to estimate CFO

$$y_1(t) = x(t)e^{-j2\pi\Delta f_c t}$$

Sample at $t = nT_s$

Symbol has N samples $\rightarrow T = NT_s$

$$y_2(t) = x(t)e^{-j2\pi\Delta f_c (t + T)}$$

OFDM CFO Estimation & Correction



- Use Preamble to estimate CFO

$$y_1[n] = x[n]e^{-j2\pi\Delta f_c nT_s}$$

Sample at $t = nT_s$

Symbol has N samples $\rightarrow T = NT_s$

$$y_2[n] = x[n]e^{-j2\pi\Delta f_c (nT_s + NT_s)}$$

- Compute:
$$A = \sum_{t=1}^N y_1[n] y_2^*[n] = \sum_{t=1}^N x[n] x^*[n] e^{j2\pi\Delta f_c NT_s}$$

$$= e^{j2\pi\Delta f_c NT_s} \sum_{t=1}^N |x[n]|^2 \quad \rightarrow \quad \Delta f_c = \frac{\angle A}{2\pi NT_s}$$

OFDM CFO Estimation & Correction



- Use Preamble to estimate CFO

$$y_1[n] = x[n]e^{-j2\pi\Delta f_c nT_s}$$

$$y_2[n] = x[n]e^{-j2\pi\Delta f_c (nT_s + NT_s)}$$

- Compute:
$$A = \sum_{t=1}^N y_1[n]y_2^*[n] \quad \longrightarrow \quad \Delta f_c = \frac{\angle A}{2\pi nT_s}$$

- Correct CFO:
$$y[n] \times e^{j2\pi\Delta f_c nT_s}$$

OFDM Channel Estimation



- Use Preamble to estimate the channel

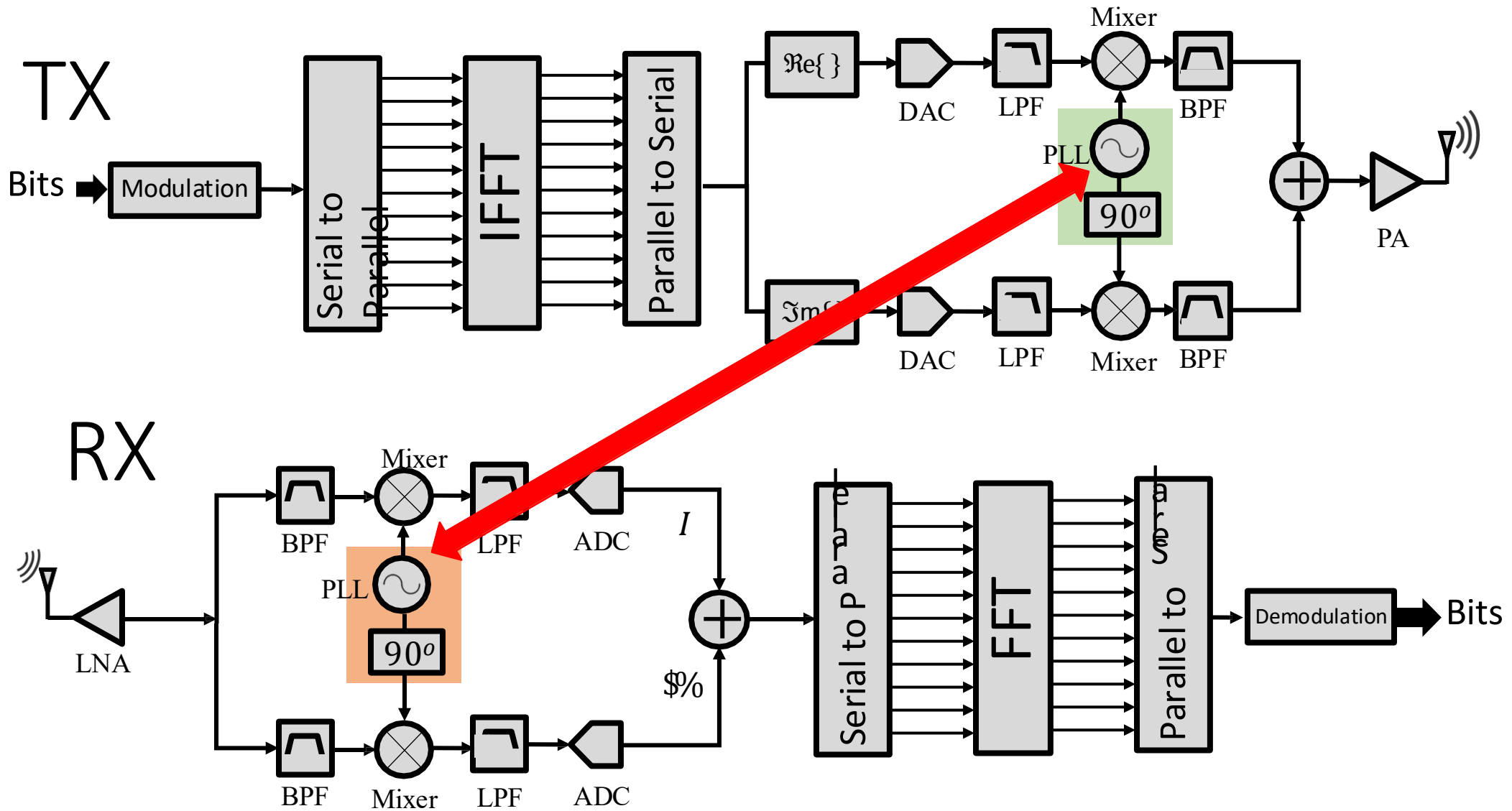
$$y(t) = h(t) * x(t) \leftrightarrow Y(f) = H(f) X(f)$$

- Send $X(f)$: $-1, +1, -1, -1, -1, +1, \dots$
- Receive: $-H(1), H(2), -H(3), -H(4), -H(5), H(6), \dots$

- Estimate: $\tilde{H}(f) = \frac{Y(f)}{X(f)}$

- Use two preambles to average noise: $\tilde{H}(f) = \frac{Y_1(f) + Y_2(f)}{2 X(f)}$

OFDM: Orthogonal Frequency Division Multiplexing



Phase Tracking

So Far: Estimated and Corrected For Coarse Value of CFO

- Residual CFO:

$$y(t) = h(t) * x(t) e^{j2\pi\Delta f_c t} + v(t)$$

$$\Delta f_c = df_c + \delta f_c$$

Coarse CFO

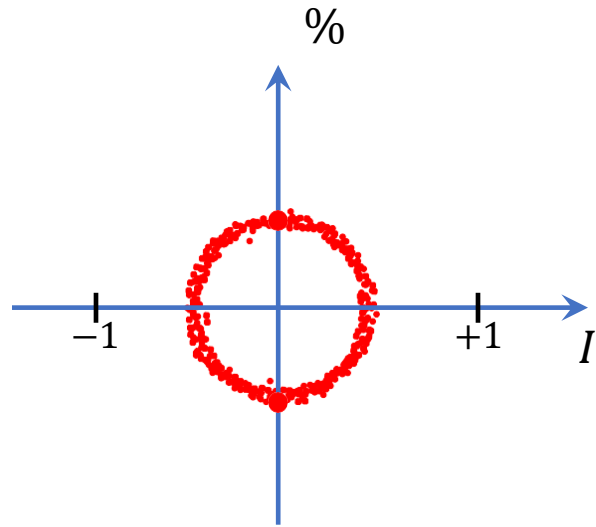
Residual CFO

We estimated and corrected for coarse CFO!

Even small residual can accumulate over time
to create large phase: $e^{-j2\pi\delta f_c t}$

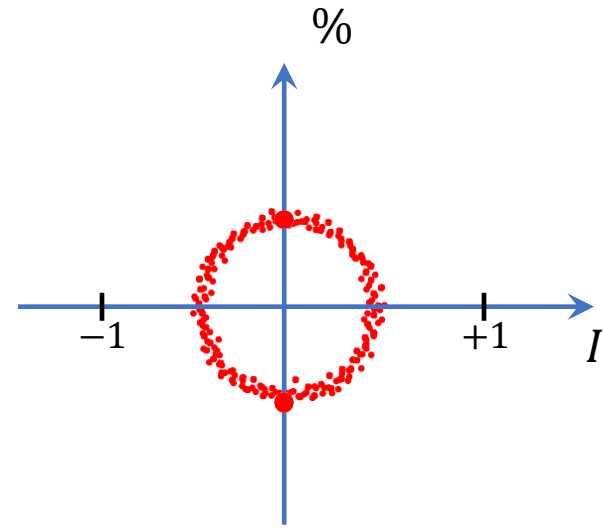
Need to track the phase

Phase Tracking



$$h x(t - r) e^{j2\pi\Delta f_c t} + v(t)$$

Without Course CFO Correction



$$h x(t - r) e^{j2\pi\delta f_c t} + v(t)$$

Without Residual CFO Correction

Phase Tracking

- Residual CFO (Carrier Frequency Offset)
- Residual SFO (Sampling Frequency Offset)

Phase Tracking

- Residual CFO (Carrier Frequency Offset)
- Residual SFO (Sampling Frequency Offset)

When we sample the signal there is a residual sampling offset: $n\delta T_s$

Phase Tracking

- Residual CFO (Carrier Frequency Offset)
- Residual SFO (Sampling Frequency Offset)

When we sample the signal there is a residual sampling offset: $n\delta T_s$

$$y[n] = x[n + n\delta T_s] e^{-j2\pi\delta f_c n T_s}$$

$$Y[f] = X[f] e^{j \frac{2\pi f n \delta T_s}{N} - 2\pi\delta f_c n T_s}$$

Phase Tracking

- Residual CFO (Carrier Frequency Offset)
- Residual SFO (Sampling Frequency Offset)

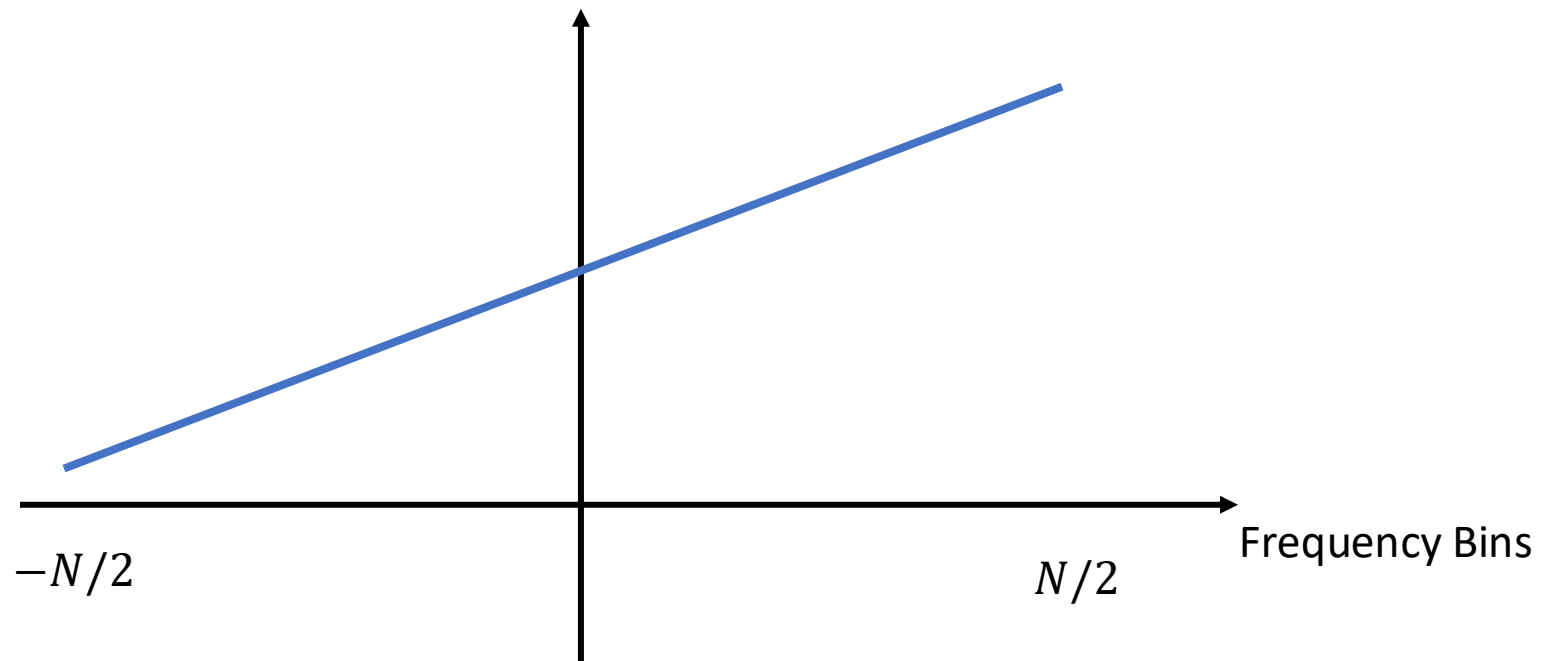
$$Y_1[f] = X_1[f] e^{j \frac{2\pi f n \delta T_s}{N} - 2\pi \delta f_c n T_s}$$

$$Y_2[f] = X_2[f] e^{j \frac{2\pi f (n+N+CP) \delta T_s}{N} - 2\pi \delta f_c (n+N+CP) T_s}$$

Phase accumulation: $\Delta\phi = 2\pi f \frac{(N+CP)\delta T_s}{N} - 2\pi \delta f_c (N+CP) T_s$

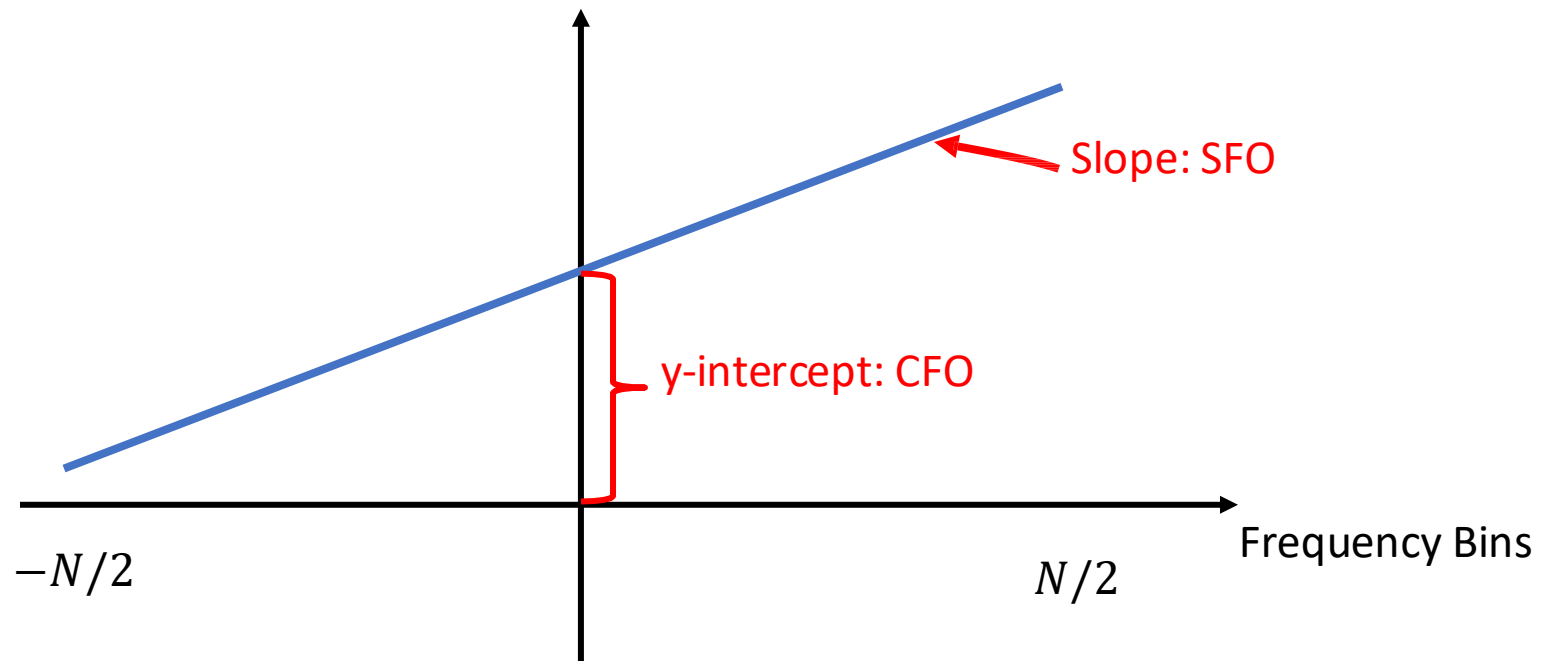
Phase Tracking

- Residual CFO (Carrier Frequency Offset)
- Residual SFO (Sampling Frequency Offset)
- Phase accumulation: $\Delta\phi = 2\pi f \frac{(N + CP)\delta T_s}{N} - 2\pi\delta f_c(N + CP)T_s$



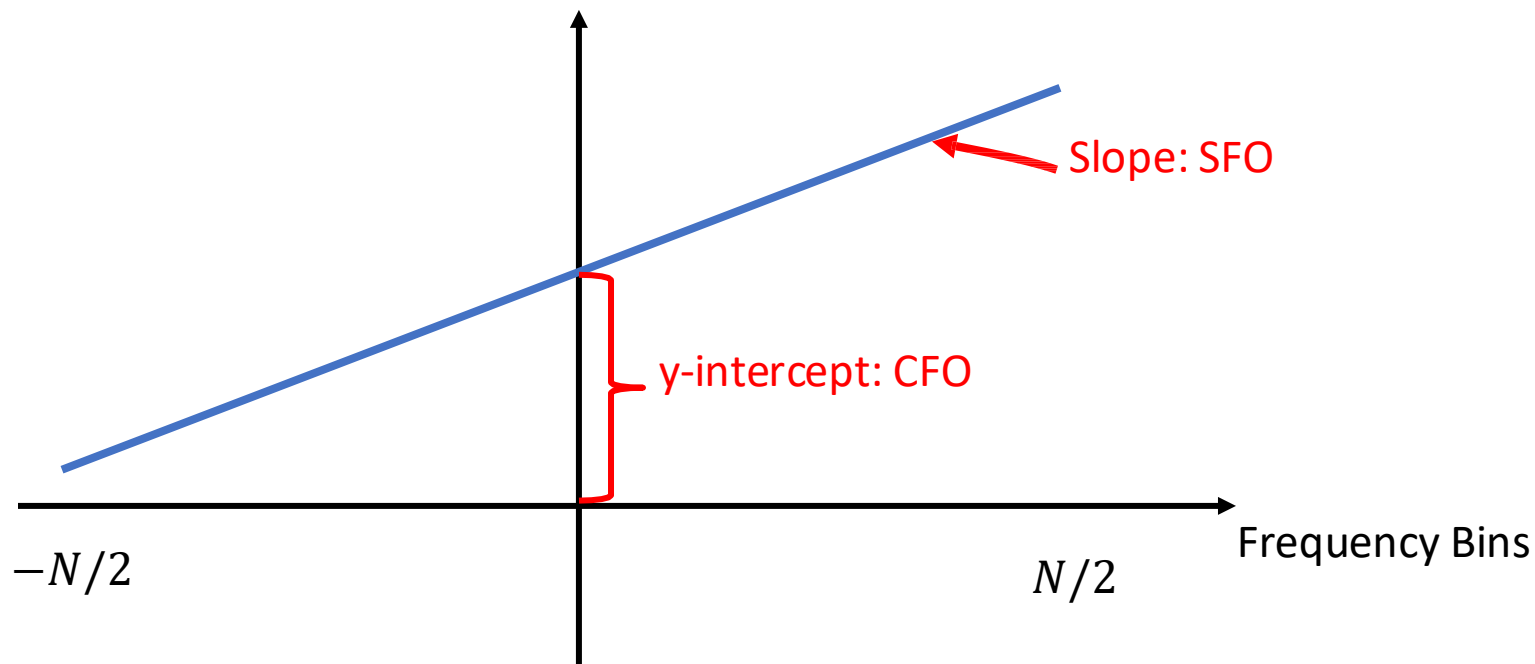
Phase Tracking

- Residual CFO (Carrier Frequency Offset)
- Residual SFO (Sampling Frequency Offset)
- Phase accumulation: $\Delta\phi = 2\pi f \frac{(N + CP)\delta T_s}{N} - 2\pi\delta f_c(N + CP)T_s$



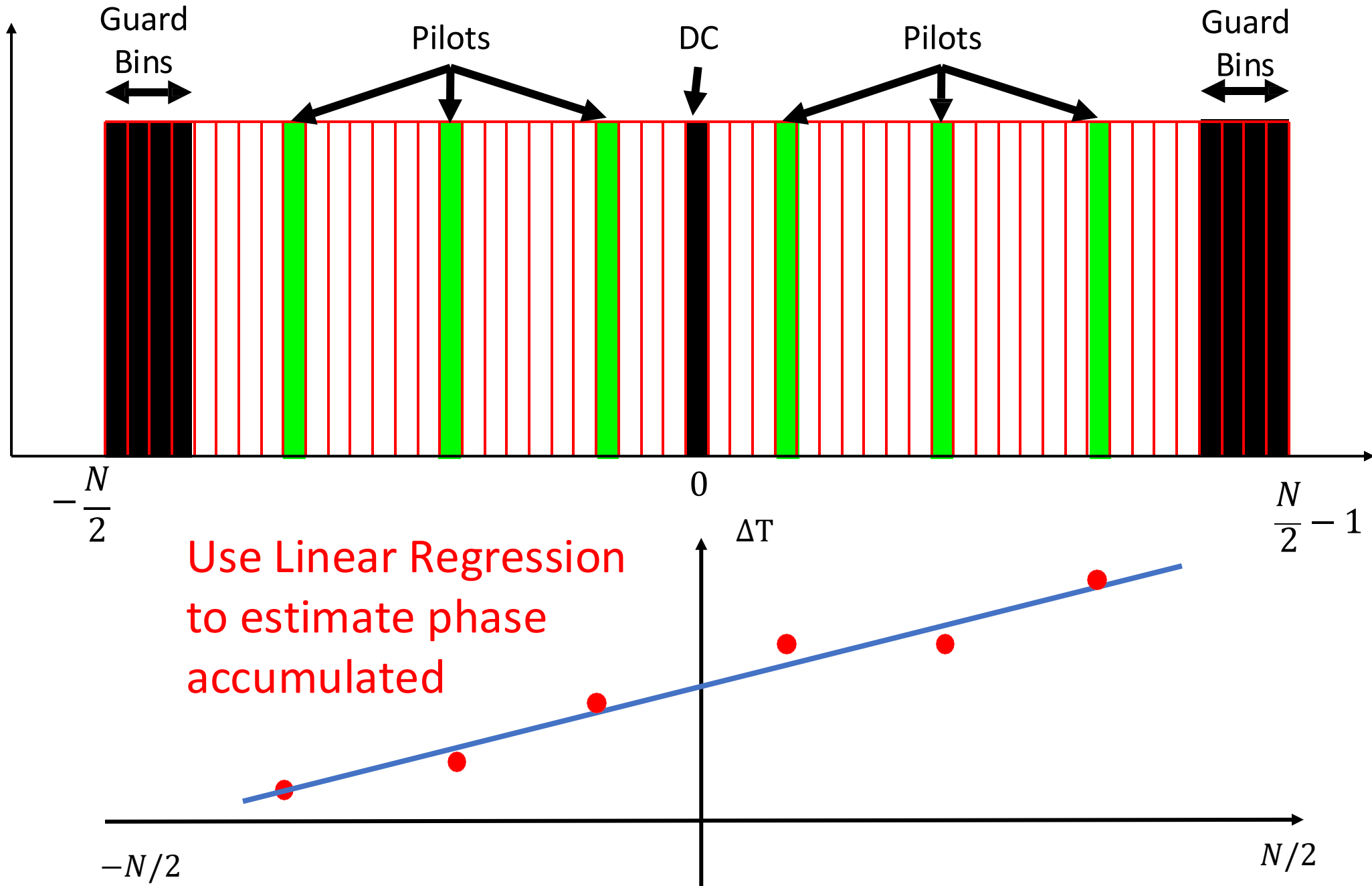
OFDM Phase Tracking

- Phase accumulation: $\Delta\phi = 2\pi f \frac{(N + CP)\delta T_s}{N} - 2\pi\delta f_c(N + CP)T_s$



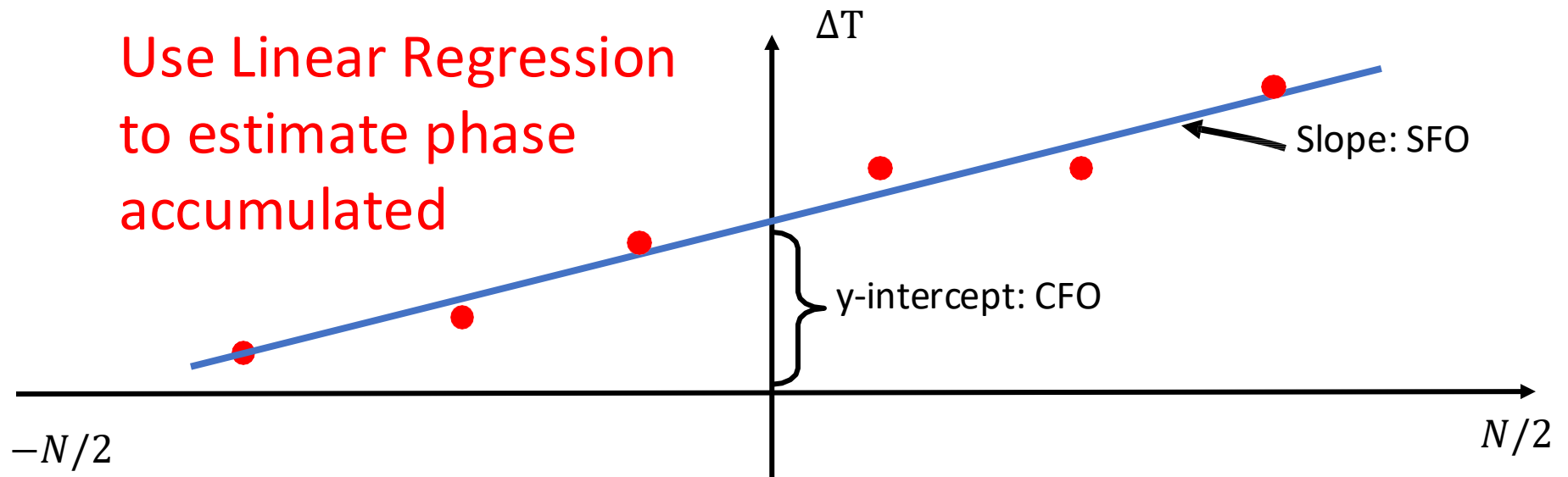
- Sufficient to estimate slope & y-intercept to know the phase accumulated for all subcarriers.
- Use only few subcarriers as pilots & send known bits in them.

OFDM Phase Tracking



OFDM Phase Tracking

Use Linear Regression
to estimate phase
accumulated

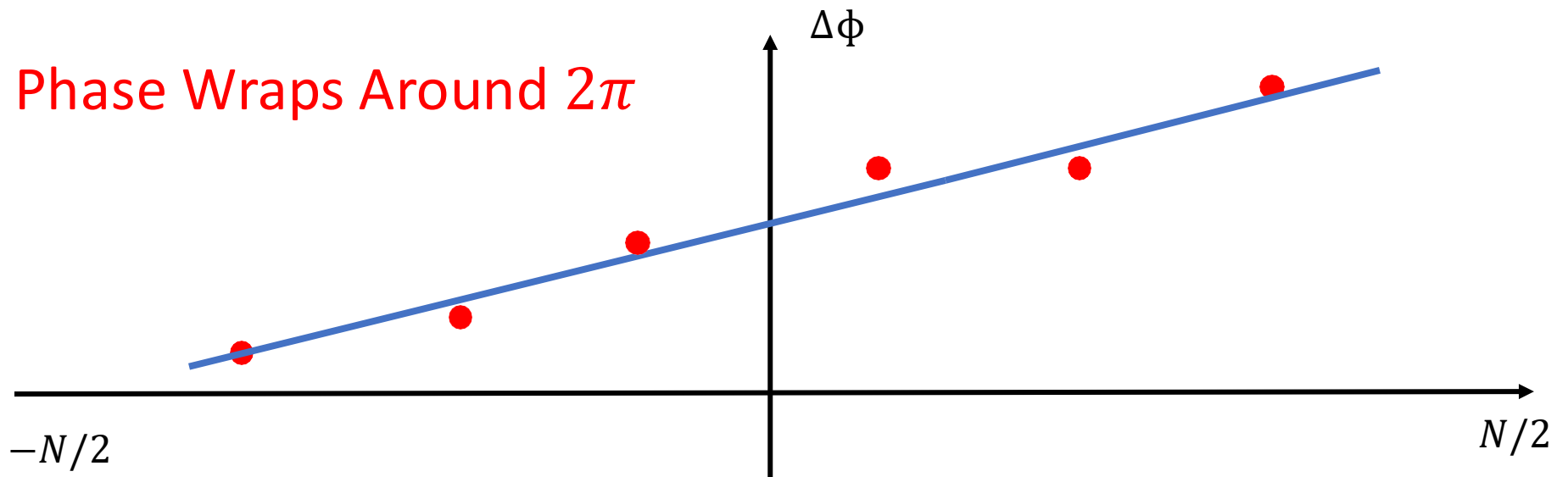


$$\text{Slope} = 2\pi \frac{(N + CP)\delta T_s}{N}$$

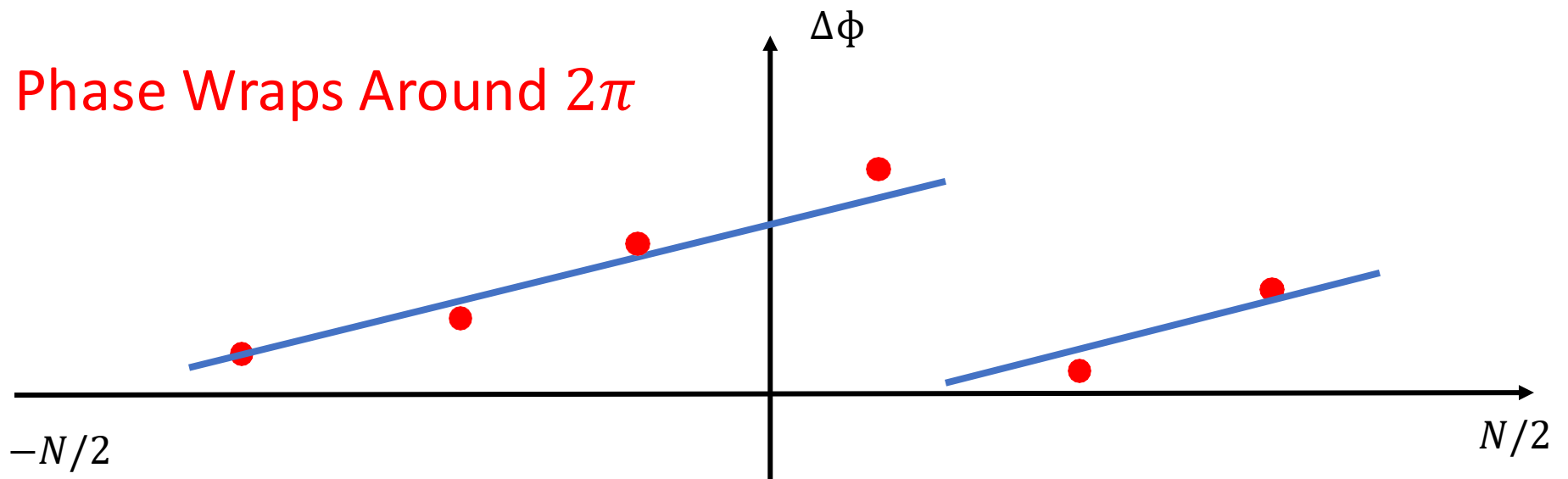
$$Y - \text{intercept} = -2\pi\delta f_c(N + CP)T_s$$

$$\Delta\phi = 2\pi f \frac{(N + CP)\delta T_s}{N} - 2\pi\delta f_c(N + CP)T_s$$

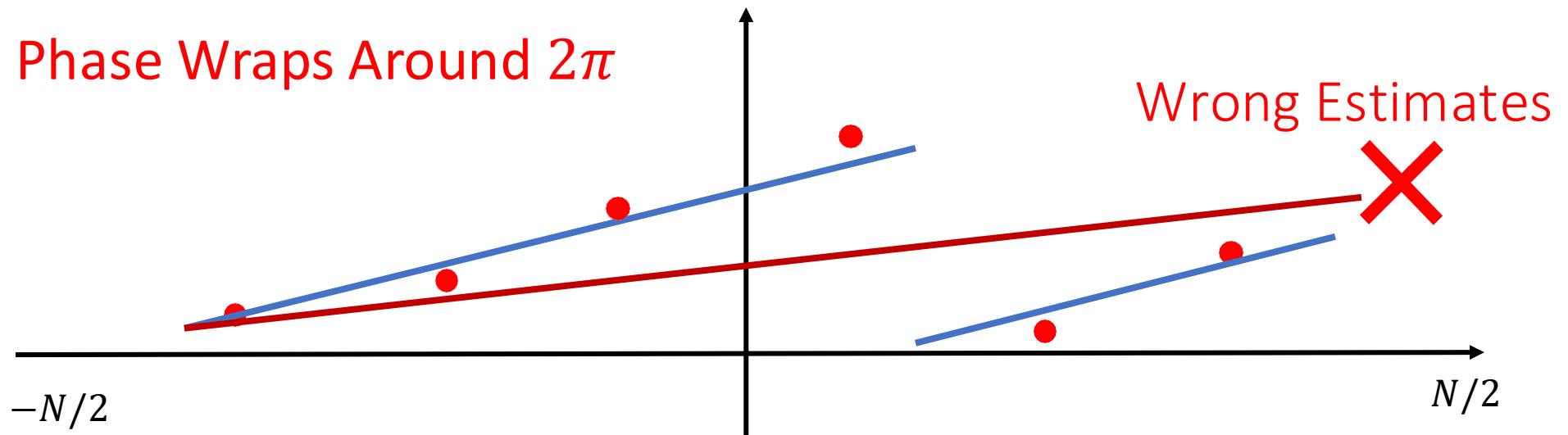
Be Careful!



Be Careful!



Be Careful!

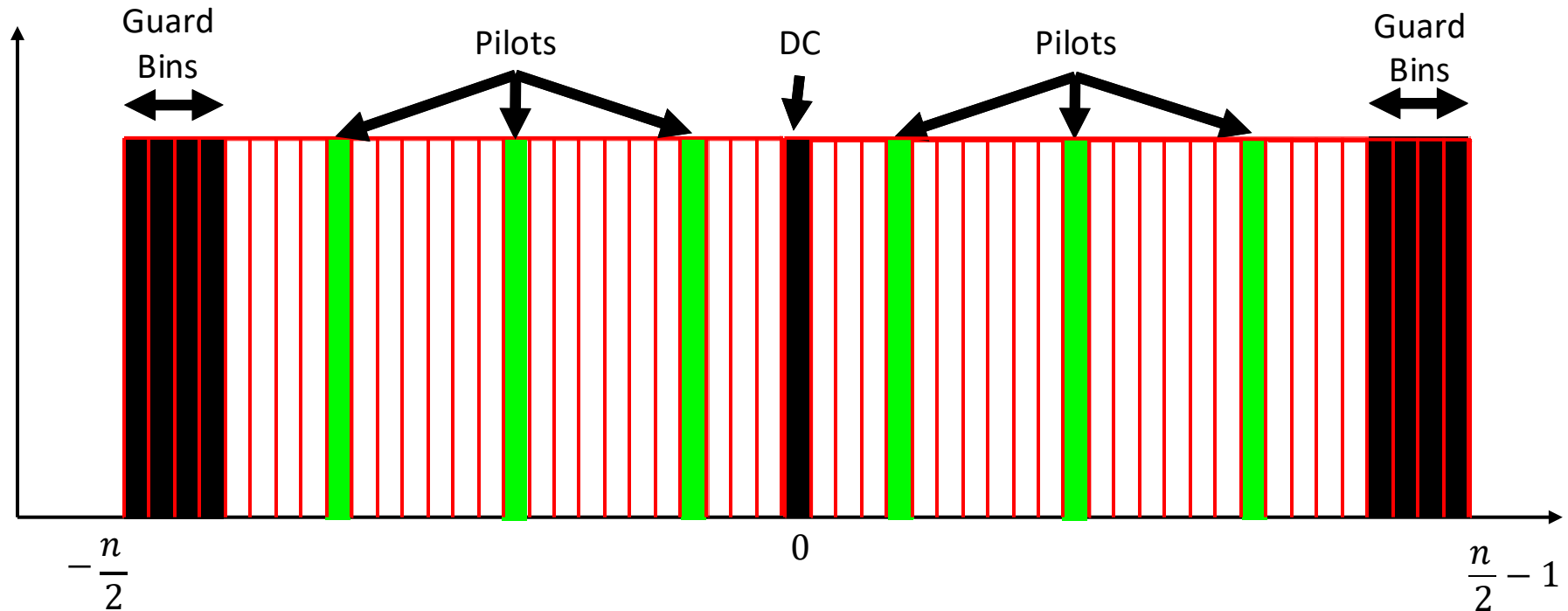


Need to unwrap phase mod 2π first! Use `unwrap()` function in MATLAB

When estimating slope and y-intercept, equation assumes subcarriers index f is between $-n/2$ and $n/2 - 1$

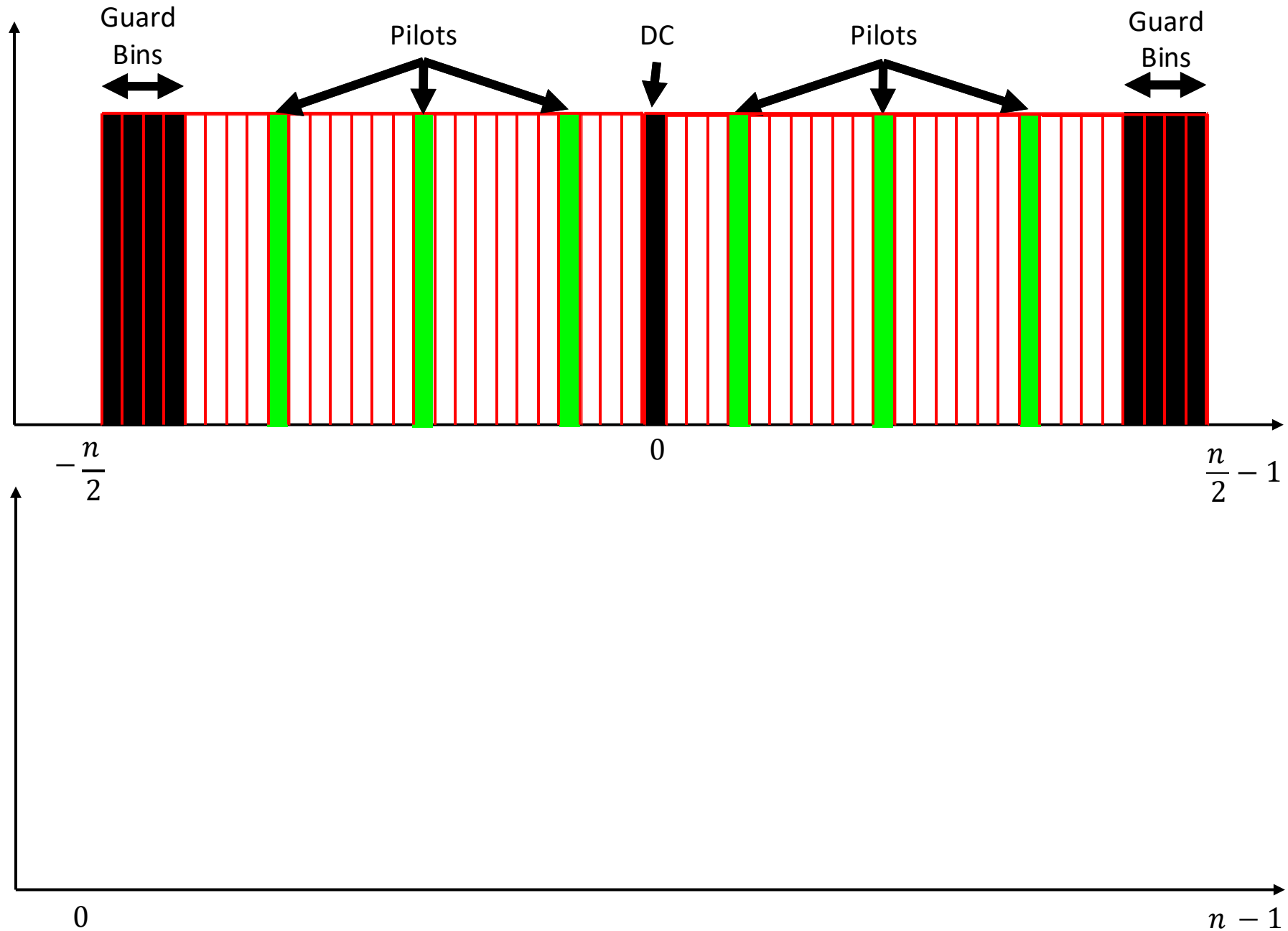
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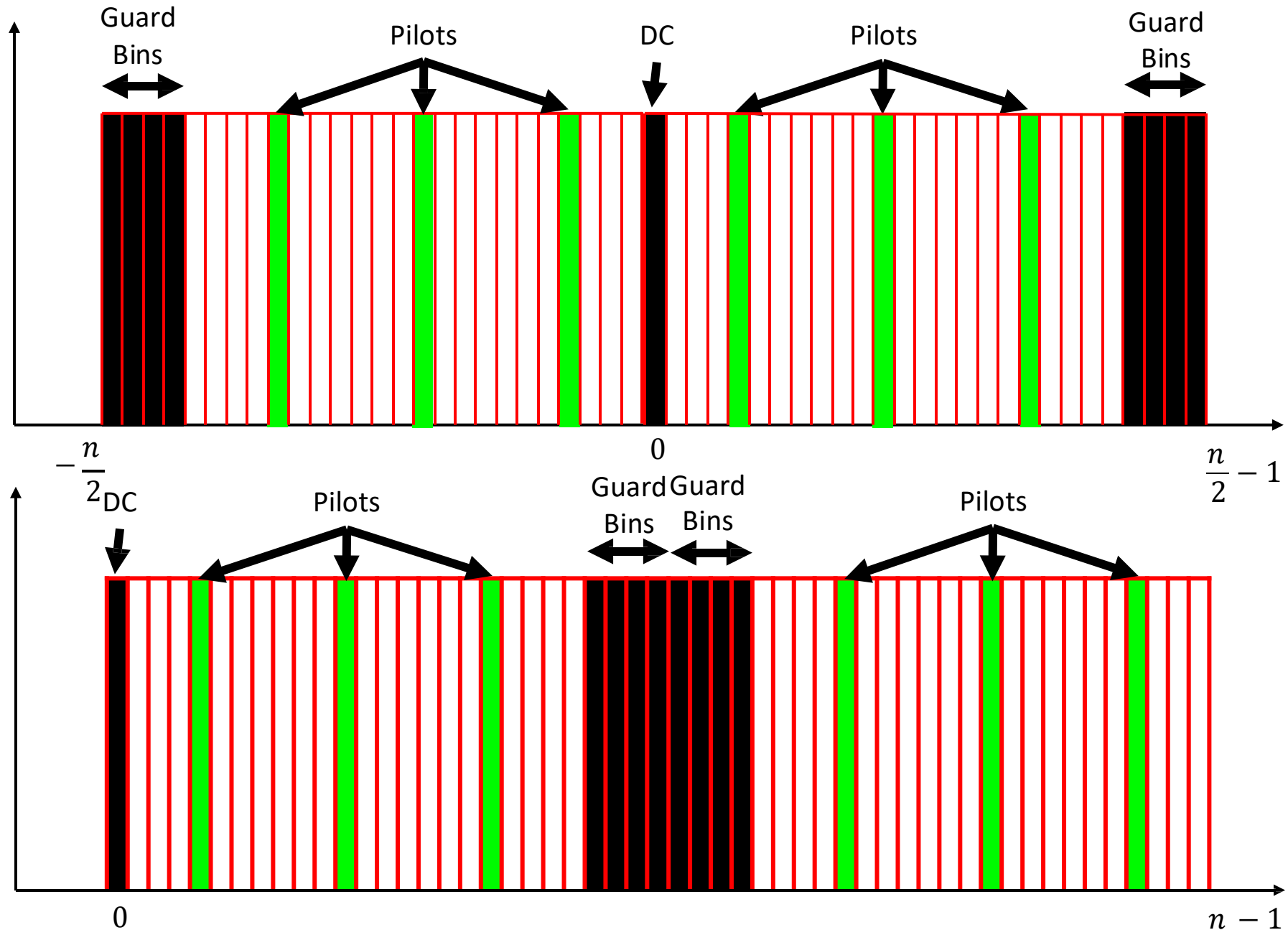
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When estimating slope and y-intercept, equation assumes subcarriers index f is between $-n/2$ and $n/2 - 1$



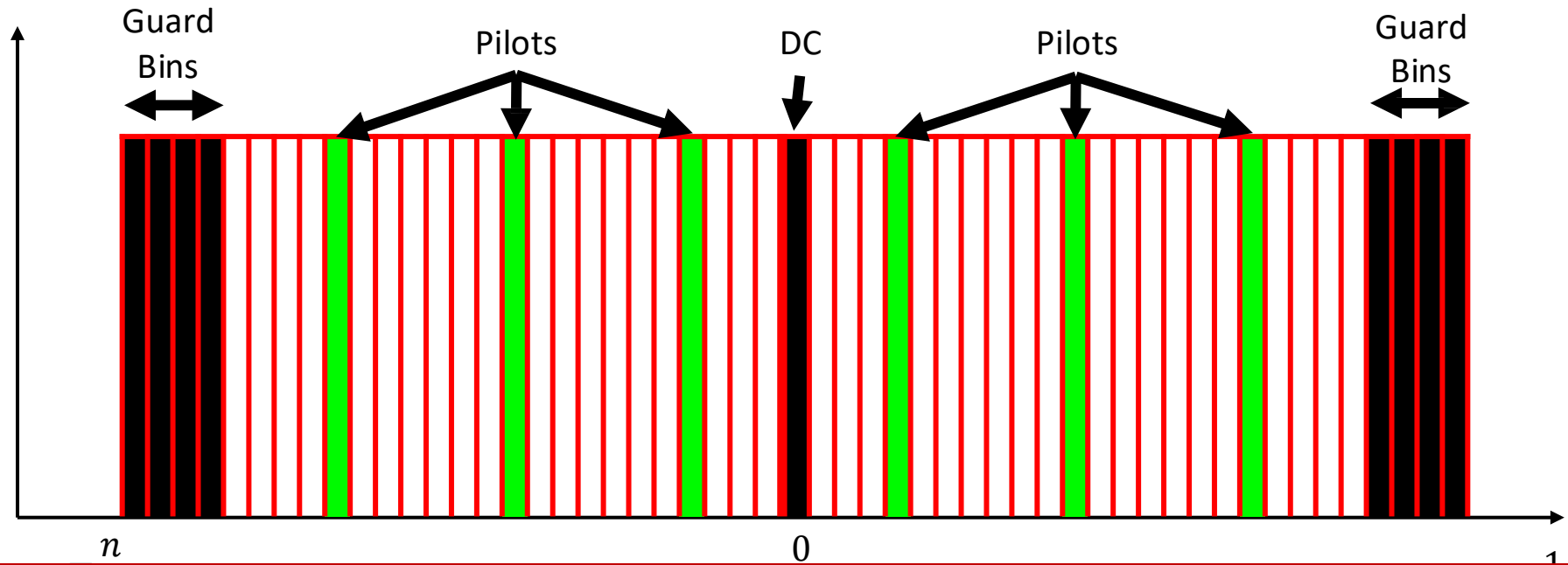
Be Careful!

When estimating slope and y-intercept, equation assumes subcarriers index k is between $-\frac{n}{2}$ and $\frac{n}{2} - 1$

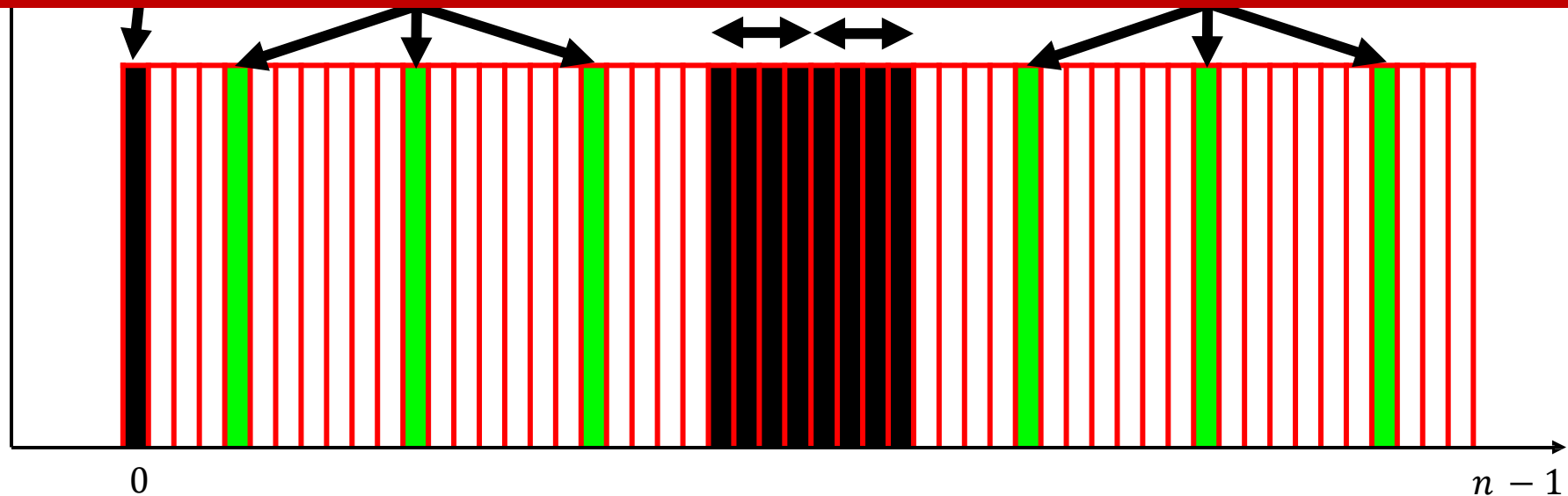


Be Careful!

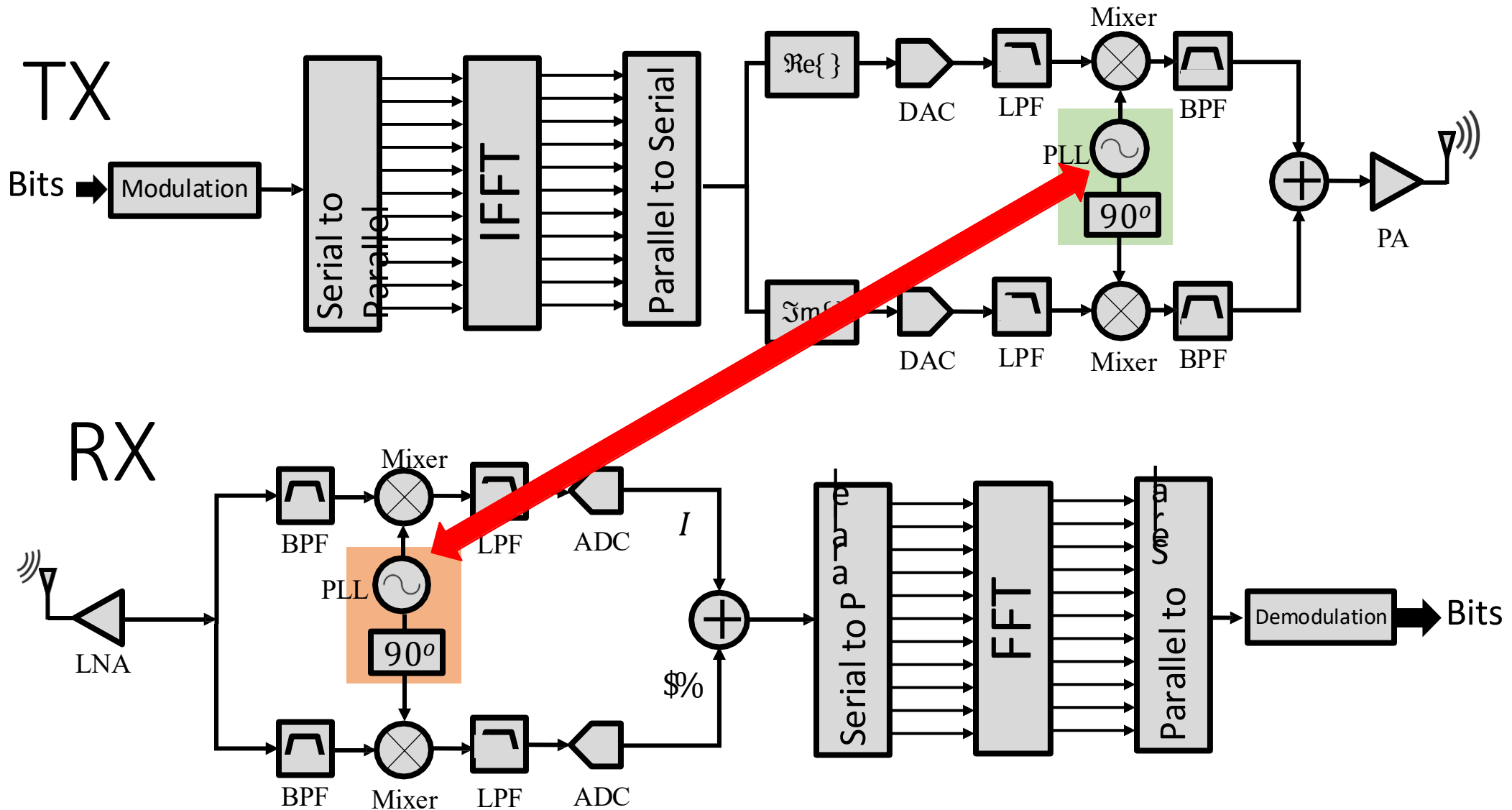
When estimating slope and y-intercept, equation assumes subcarriers index +is between $-n/2$ and $n/2 - 1$



Need to `fftshift()` function in MATLAB + provided functions to convert indexing.



OFDM: Orthogonal Frequency Division Multiplexing



OFDM: Putting it Together

At TX:

- Create preamble symbol from training sequence (Uses BPSK)
- Repeat preamble symbol:
 - ~10 times STS for coarse packet detection
 - 2 times LTS for fine packet detection
 - 2 times LTS for CFO estimation
 - 2 times LTS for channel estimation
 - Add CP for LTS preamble
- Create data symbol from:
 - Data bits (Uses BPSK, QPSK, M-QAM)
 - Pilot bits (Uses BPSK)
- Add cyclic prefix to data symbols.

OFDM: Putting it Together

At RX (HW 4):

- Detect beginning of packet.
- Estimate & correct for CFO.
- Jump $\approx CP - 4$ samples into symbol to avoid ISI
- Estimate the channel.
- For each subsequent data symbol:
 - Remove CP
 - Take FFT of Size N
 - Correct for channel by dividing with $\tilde{H}(f)$
 - Use linear regression to estimate residual CFO and SFO with pilots
 - Estimate accumulated phase $\Delta\phi_f$ for each frequency bin
 - Correct for residual CFO and SFO
 - Decode Bits